

DATA VORTEX 2026 · ROUND 1 PHASE 2

Explanation of logic

Team SE7EN · E3, M2, H4

Why these three, how engagement is defined, and what we refused to do

1. Why these three questions

The questionnaire says pick one from each band. We did not pick at random. The three we kept answer one actual question about the Social Engine: does audience size explain engagement, and if not, what does?

Easy — E3, not E1

E1 is a COUNT(*) GROUP BY platform. Facebook has 2,074 posts, YouTube 2,073. Calling Facebook “the most popular platform” off a one-post gap on 10,216 labelled rows is noise. E3 asks the thing you would actually ship: once a post is up, which network gets the work done. It also forces the NULL question, because AVG(likes) and AVG(likes + shares + comments) do not use the same denominator.

Medium — M2, not M1

M1 ranks locations by total engagement. Los Angeles leads, and Los Angeles also has the most posts. That ranking is mostly volume. M2 is a hypothesis test the brief is built for: split the authors at 25,000 followers and see whether the bigger accounts actually earn more per post. They do not. A two-row result that looks “too simple” is the result.

Hard — H4, not H1 or H3

H1 asks for users whose average engagement is more than twice the overall average. The overall mean is 4,001.99, so the bar is 8,003.97. A post cannot exceed 5,000 likes + 2,000 shares + 1,000 comments = 8,000, and the highest post in the extract is 7,893. H1 is empty by construction. H3 has the same bound against each platform mean. We ran both to confirm, then left them. H4 is the item the questionnaire itself flags as needing more than one layer: build a user total, find the top 10% of those totals, then keep the accounts with fewer than 5,000 followers.

H5 would have been a rerun of Phase 1 on the corrupted file. The judges already have that work. Phase 2 is the analytical core, so we stayed on the cleaned tables.

2. One definition of engagement, used everywhere

```
likes + shares + comments    -- NULL when likes IS NULL
```

Shares and comments are complete. Likes are missing on 1,814 posts (15.1%). In SQL, NULL + 400 + 50 is NULL, and AVG() skips NULL. That is what we want. Filling likes with 0 would say “nobody liked this” for rows where the instrument failed. Filling with the median (2,498) would pin 1,814 posts on a single value and shrink the spread the later questions are trying to read. Phase 1 already showed the missingness is MCAR, so dropping those rows from the mean is unbiased.

A related trap: AVG(likes) + AVG(shares) + AVG(comments) is not equal to AVG(likes + shares + comments). Likes runs over 10,186 rows; shares and comments run over 12,000 (or over the platform subset). On Instagram the sum of the three averages is 4,041.57; the average of the sums is 4,040.02. The question asks for average total engagement, so we take the second one.

3. E3 — walkthrough

- WHERE platform IS NOT NULL. The 1,784 blank platforms have no group. We did not dump them into an “Unknown” bucket; that would be a different question.
- COUNT(*) is volume. COUNT(likes) is the denominator of avg_likes and of avg_total_engagement. Both are printed so the two denominators are visible.
- ROUND(..., 2) is display. Ordering uses the rounded alias in MySQL 8, which is enough to separate Instagram (4040.02) from YouTube (4031.84).
- Row 1 is the answer: Instagram.

We considered restricting to posts where likes, shares and comments are all present. Shares and comments never go missing, so that filter is just likes IS NOT NULL, which AVG already does for the total.

4. M2 — walkthrough

- The CASE belongs in a CTE so the GROUP BY is a single column, not a copy-pasted expression.
- The grain is the post. A 22-post account contributes 22 times. Averaging first at user grain and then averaging those averages would give every account equal weight and answer a different question (“do high-follower authors have a higher personal mean?”).
- The cut is the one in the brief: $\geq 25,000$ vs $< 25,000$. We did not move it to the median (24,742) even though that would have balanced the groups more neatly.
- vs_overall_avg subtracts the table-wide post mean (4,001.99) so the screenshot shows -1.07 and $+1.03$ instead of two numbers that look accidentally identical.

Join is INNER JOIN. Every post has a user, every user has at least one post, so INNER and LEFT produce the same 12,000 rows. We still write INNER because that is the relationship we mean.

5. H4 — walkthrough

Three layers, matching the note on the question (“this requires multiple levels of analysis”).

- Layer 1. One row per user: post_count, posts_with_engagement, AVG of the engagement expression, SUM of the same expression. SUM of an all-NULL set is NULL. That happens for two users. They cannot be in a top 10% of total engagement.
- Layer 2. NTILE(10) OVER (ORDER BY total_engagement DESC). 1,500 users, ten buckets of 150. Decile 1 is the top 10%. We did not write WHERE total_engagement > 41882, because that number is a property of this extract and the brief disqualifies hardcoded outputs.
- Layer 3. follower_count < 5000 inside that decile. Fifteen rows come back. ORDER BY total_engagement DESC so the most extreme small account is on top.

PERCENT_RANK() OVER (ORDER BY total_engagement) is multiplied by 100 and printed. 100 would be the single highest total in the 1,500. user_fgjkkrie sits at 99.73 with 2,211 followers, which is the point of the question. The percentile is a reading aid; the filter is NTILE.

NTILE vs CUME_DIST vs “rank <= 150”: on this extract all three keep the same 150 users in the head of the list, and the same 15 after the follower cut. We kept NTILE because “top 10%” is a decile in plain language.

6. Things we refused to do

- `COALESCE(likes, 0)` or `COALESCE(platform, 'Unknown')`. That is fabrication under the Round 1 rulebook, and it would change every mean in this pack.
- Dropping incomplete posts up front. Only 61% of rows are complete. The missingness is MCAR; throwing 39% of the table away buys nothing.
- Hardcoding “Instagram” or a percentile cutoff into the `SELECT`. The statements have to recompute.
- Picking H1 or H3 and then writing a paragraph about an empty grid. Empty is the correct answer to those two, but it is not a useful Phase 2 submission.